

Image-to-SMILES Representative Panel

PANEL_IMAGE_TO_SMILES — 16 rows

2026-05-24

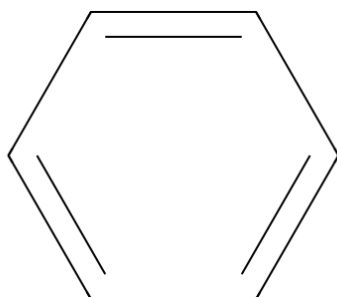
Abstract

This panel is the 16-row representative slice of `tests/ketcher/image-to-smiles/manifest.jsonl` used as a quick regression sweep for the `ketcher-image-rebuild` skill. Each row probes a distinct failure surface — baseline aromatic, drawn-H + charges, wedge CIP, *E/Z* encoding, adversarial filename, refusal, α/β anomer, multi-charge, macrocycle topology, and the tier-2 dense-stereo cluster (PLAN-a004-class-robustness). Together they test reconstruction capability across the full complexity range from 6-heavy aromatic to 62-heavy fused taxane (paclitaxel) without paying for the 98-row full image manifest. Triggered by `Run image-to-smiles panel per the runbook` or `/test image-to-smiles-panel`. Panel definition (id list + trigger phrase) lives in `tests/AGENT_RUNBOOK.md §Panels`; this PDF is the typeset companion.

Panel members

I001 Benzene

shard: S9-clean



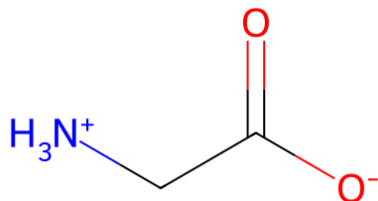
Probes: Baseline aromatic round-trip — smallest non-trivial fixture.

Why: Kekulé $\rightarrow$ aromatize $\rightarrow$ export pipeline on a perfectly symmetric arene. If this fails, nothing downstream works.

Signature: heavy=6, rings=1 aromatic, wedges=0, charges=0, drawn_H=0.

I007 Glycine zwitterion

shard: S9-clean



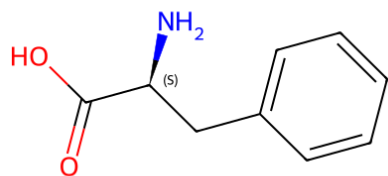
Probes: Drawn-H plus formal-charge preservation under canonicalization.

Why: Silent neutralization on export — RDKit canonicalization erases the zwitterion, so any pipeline that round-trips through canonical loses the N⁺/O⁻ pair.

Signature: heavy=5, rings=0, charges=2 (N⁺/O⁻), drawn_H on N.

I008 L-phenylalanine

shard: S9-clean



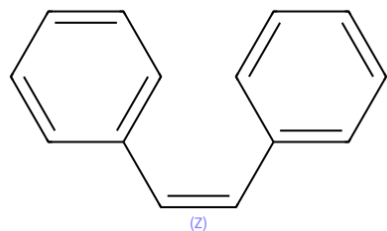
Probes: Single-wedge CIP at C_α ; “(S)” label printed at the centre.

Why: May 2026 `set_wedge_bond` L $\leftrightarrow$ D inversion — when the chiral atom is not made `bond.begin`, exported CIP silently flips L to D.

Signature: heavy=12, rings=1 aromatic, wedges=1, charges=0.

I009 cis-stilbene

shard: S9-clean



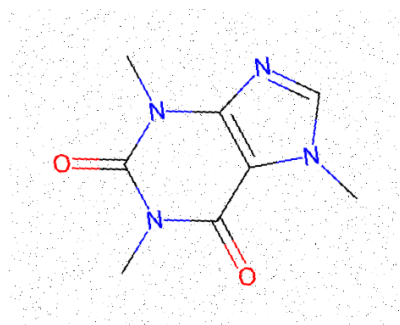
Probes: *E/Z* encoding on the central C=C (both phenyls same side, “(Z)” label visible).

Why: First-guess defaults to trans-stilbene from rote memory; the row catches that single-shot inversion.

Signature: heavy=14, rings=2 aromatic, wedges=0, `cis_trans`=1.

I011 Caffeine

shard: S9b-noisy



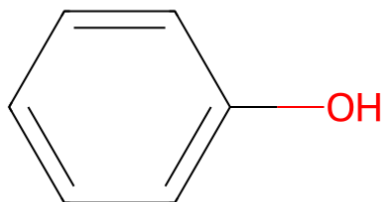
Probes: Filename-leakage resistance — file is called `rotated_not_aspirin.png`, drawn molecule is caffeine.

Why: Any pipeline that short-circuits to the filename token rather than transcribing the pixels exports aspirin instead.

Signature: heavy=14, rings=2 (6+5 fused purine), 2 C=O, 3 N-methyls.

I017 Phenol

shard: S9c-adversarial



Probes: Second filename-leakage fixture (different compound pair) — file says toluene, drawn molecule is phenol.

Why: Generalization check: confirms the filename-refusal behaviour does not depend on the specific caffeine/aspirin pairing in I011.

Signature: heavy=7, rings=1 aromatic, OH at one ring position.

N001 Negative control — reaction arrow only

shard: S10-negative



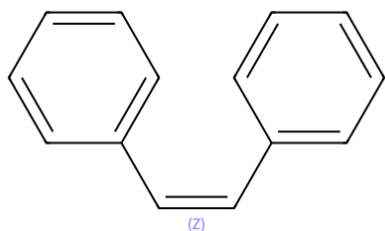
Probes: Refusal — image is a bare arrow on blank background; no atoms, bonds, rings, or labels.

Why: “Hallucinate a SMILES anyway” failure mode. Skill must emit “Recognition failed.” and *must not* call `load_smiles`, `load_canonical`, or `build_from_graph`.

Signature: No chemistry. Trace must contain `handle_recognition_failure_without_invention`.

RT001 cis-stilbene (round-trip regression)

shard: S13-roundtrip



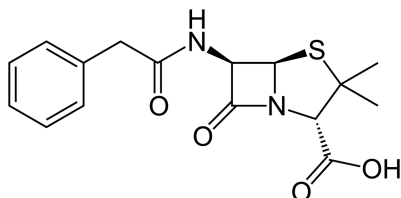
Probes: Stage R readback-loop catches the same trans→cis flip as I009, but under the `image_roundtrip_evaluator` grading mode.

Why: Verifies the readback iteration corrects a wrong first guess *before* export, not only the one-shot build path.

Signature: Same source image as I009: `heavy=14`, `cis_trans=1`.

A002 Penicillin G

shard: S11-academic



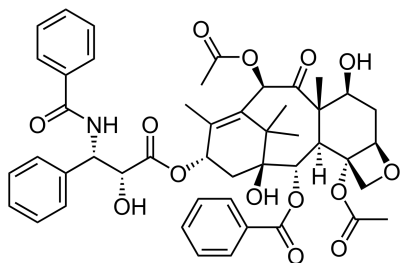
Probes: Rare fused topology — 4-membered β -lactam plus 5-membered thiazolidine, joined at both a C and an N.

Why: 4+5 ring fusion is unique in the panel; stresses the translator’s ring perception. Also catches stereo at three centres (2S, 5R, 6R).

Signature: `heavy=23`, `rings=3` (4+5+6arom), `wedges=3`, `charges=0`, `drawn_H=2`.

A004 Paclitaxel (taxol)

shard: S11-academic



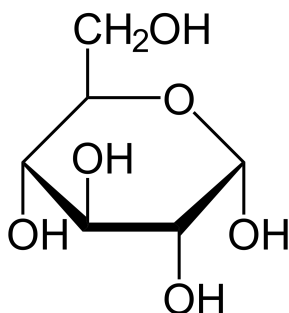
Probes: Original dense-stereo ceiling — full paclitaxel with taxane core (8+6+6 plus 4-membered oxetane), benzoate ester, N-benzoyl phenylisoserine side chain.

Why: Anchors what “works” at K=11 stereocenters under Mode B parity-transfer. Well-behaved baseline for the tier-2 cluster below; passed in prior session.

Signature: `heavy=62`, `rings=7` (3 aromatic), `wedges=11`, `charges=0`.

A005 α -D-glucopyranose

shard: S11-academic



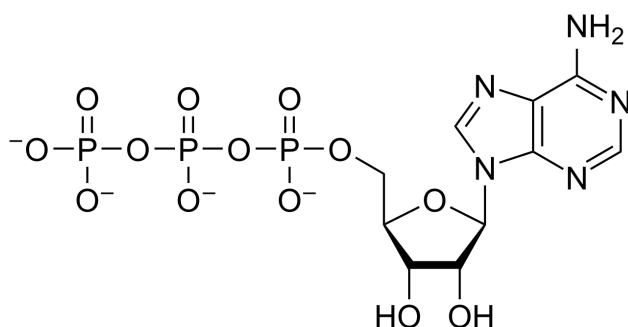
Probes: Anomeric stereo — α (–OH axial, drawn down) vs β (equatorial, drawn up); Haworth projection.

Why: May 2026 regression: expected SMILES was the β anomer despite the image being α . Tests W2 **verticalSense** Haworth-projection parity-transfer.

Signature: heavy=12, rings=1 (6 non-arom with one ring O), wedges=5, charges=0.

A006 ATP tetraanion

shard: S11-academic



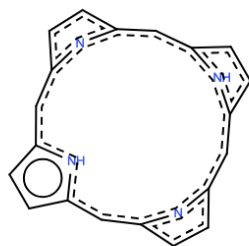
Probes: Multi-charge preservation — adenine purine plus ribose furanose plus triphosphate with *four explicit O⁻ charges* drawn.

Why: Any pipeline that “corrects” to the neutral free acid loses the four charges. Also catches the P $_{\alpha}$ /P $_{\beta}$ spurious-stereo RDKit-canonicalization edge case.

Signature: heavy=31, rings=3 (6arom + 5arom + 5aliph), wedges=4, charges=4, drawn_H=3.

D055 Porphine

shard: S26-macrocycle



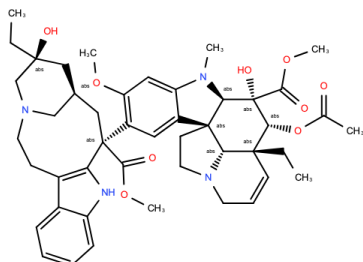
Probes: Macrocycle topology — four pyrrole rings linked by four methine bridges into a 16-membered macrocycle.

Why: Pure-topology fixture (no stereo, no charges). N-H tautomer placement — which 2 of 4 nitrogens carry H — is the discriminator.

Signature: heavy=24, rings=5 (4 $\times$ 5-arom + 16-arom), wedges=0, charges=0, drawn_H=2.

A009 Vinblastine (tier-2)

shard: S11-academic



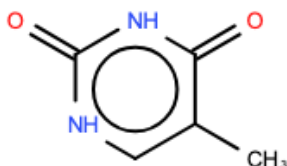
Probes: Tier-2 PLAN-a004-class-robustness — dimeric *Catharanthus* indole alkaloid joined across two indoline+indole halves.

Why: Extends the paclitaxel dense-stereo class; saddle-cluster transcription on a 9-ring scaffold.

Signature: heavy=59, rings=9 (5 $\times$ 5 + 4 $\times$ 6 + 1 $\times$ 9), K~9 stereocenters, charges=0.

A010 Thymine (image-truth tautomer)

shard: S22-nucleobase



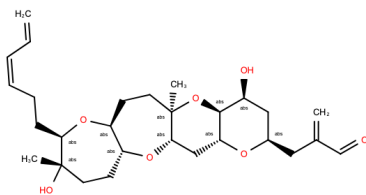
Probes: Image-truth tautomer probe — the drawn structure is the 2,4-dione lactam (two N-H, two C=O); canonical SMILES would unify with the chemically-equivalent lactim (C=N + OH) form under InChI / RDKit normalization.

Why: Isolated drawn-tautomer / drawn-C=O-position failure surface. D055 (porphine NH-tautomer) is the only other panel row probing this axis, and there the tautomer signal is confounded with macrocycle complexity. A010 isolates the tautomer probe on the smallest, cleanest fixture in the manifest — if the evaluator vision-compare step doesn't enforce drawn tautomer here, it won't enforce it anywhere.

Signature: heavy=9, rings=1 (6-mem aromatic), K=0, drawn-H=2 (two explicit N-H), wedges=0, charges=0.

A011 Hemibrevetoxin B (tier-2)

shard: S11-academic



Probes: Tier-2 PLAN-a004-class-robustness — marine polyether ladder, four trans-fused tetrahydropyran/oxepane rings plus a diene tail and an α -methylene aldehyde tail.

Why: Ladder-fusion topology slot. Failure mode in prior session was V4 coord-pin omission on the diene under `layoutPolicy:ketcher_clean_locked` ($Z \rightarrow E$ flip).

Signature: heavy=35, rings=4 ($2 \times 6 + 2 \times 7$), wedges=10, `cis_trans=1`, K=10 stereocenters.

Coverage matrix

Failure surface	Rows probing it
Baseline image-rebuild (no stereo)	I001, D055
Drawn-H + zwitterion charges	I007
Wedge CIP (R vs S)	I008, A002, A005, A004
E/Z (cis vs trans)	I009, RT001, A011
Adversarial filename (trust pixels)	I011, I017
Negative-control refusal	N001
Stage R round-trip readback loop	RT001
Rare topology (fused β -lactam, macrocycle, ladder)	A002, D055, A011
Multi-charge (formal anions)	A006
Dense-stereo ceiling ($K \geq 9$)	A004, A009, A011
Drawn-tautomer / drawn-C=O position (image-truth)	A010, D055

Trigger: Run `image-to-smiles` panel per the runbook or `/test image-to-smiles-panel`.
 Panel definition: `tests/AGENT_RUNBOOK.md §Panels`.